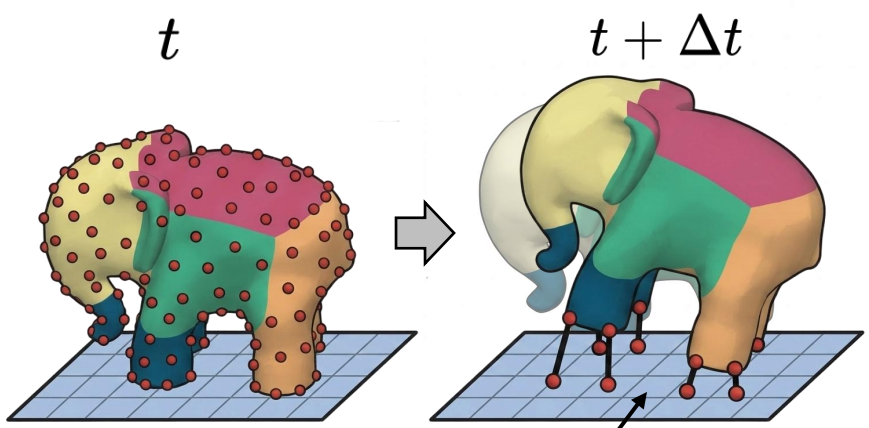
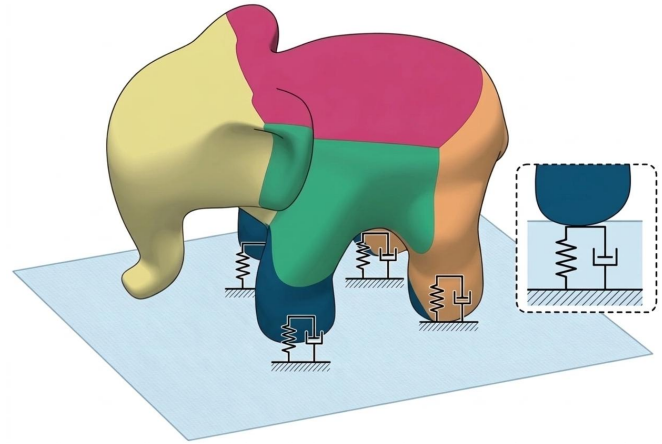
Assumption 1: Fixed Contact Set



Contacts remain unchanged over a short time interval (Sticking).

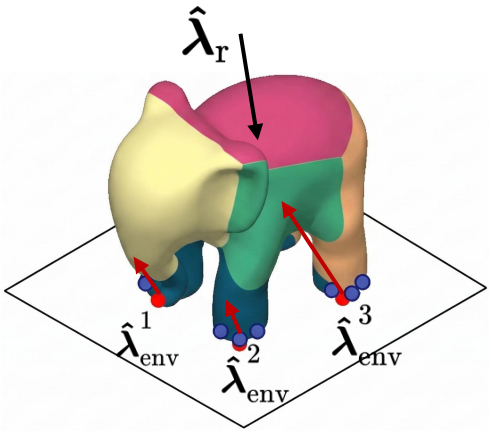
Assumption 2: Multi-Contact Decoupling



Independent Spring-Damper System

Neglecting the coupling among environment-object contacts

Surrogate Contact Model



$$\hat{\boldsymbol{p}}_r \in \mathcal{I}_{\text{avail}},\ \hat{\boldsymbol{\lambda}}_{\mathrm{r}} \in \mathcal{K}$$

$$\hat{\boldsymbol{\lambda}}_{\mathrm{env},i} = \Pi^{T/h}_{\mathcal{K}_i}\left(-\boldsymbol{D}^{-1}\left(\tilde{\boldsymbol{J}}_{\mathrm{env},i}\boldsymbol{M}_o^{-1}\tilde{\boldsymbol{b}}_o + \tilde{\boldsymbol{\phi}}_i/h^2\right)\right).$$

Gauss-Seidel

$$f_{\mathrm{SCM}}: \begin{cases} \hat{\boldsymbol{v}}^+ = h\boldsymbol{M}_o^{-1}\tilde{\boldsymbol{b}}_o + h\boldsymbol{M}_o^{-1}\sum_{i=1}^{n_{\mathrm{env}}} \tilde{\boldsymbol{J}}^\top_{\mathrm{env},i}\hat{\boldsymbol{\lambda}}_{\mathrm{env},i}, \\ \hat{\boldsymbol{x}}_{k+1} = \boldsymbol{x}_k \oplus h\hat{\boldsymbol{v}}^+. \end{cases}$$

SCM is a lightweight approximation of the object-environment contacts. With SCM, the environment contact force is an explicit function of robot contact force.